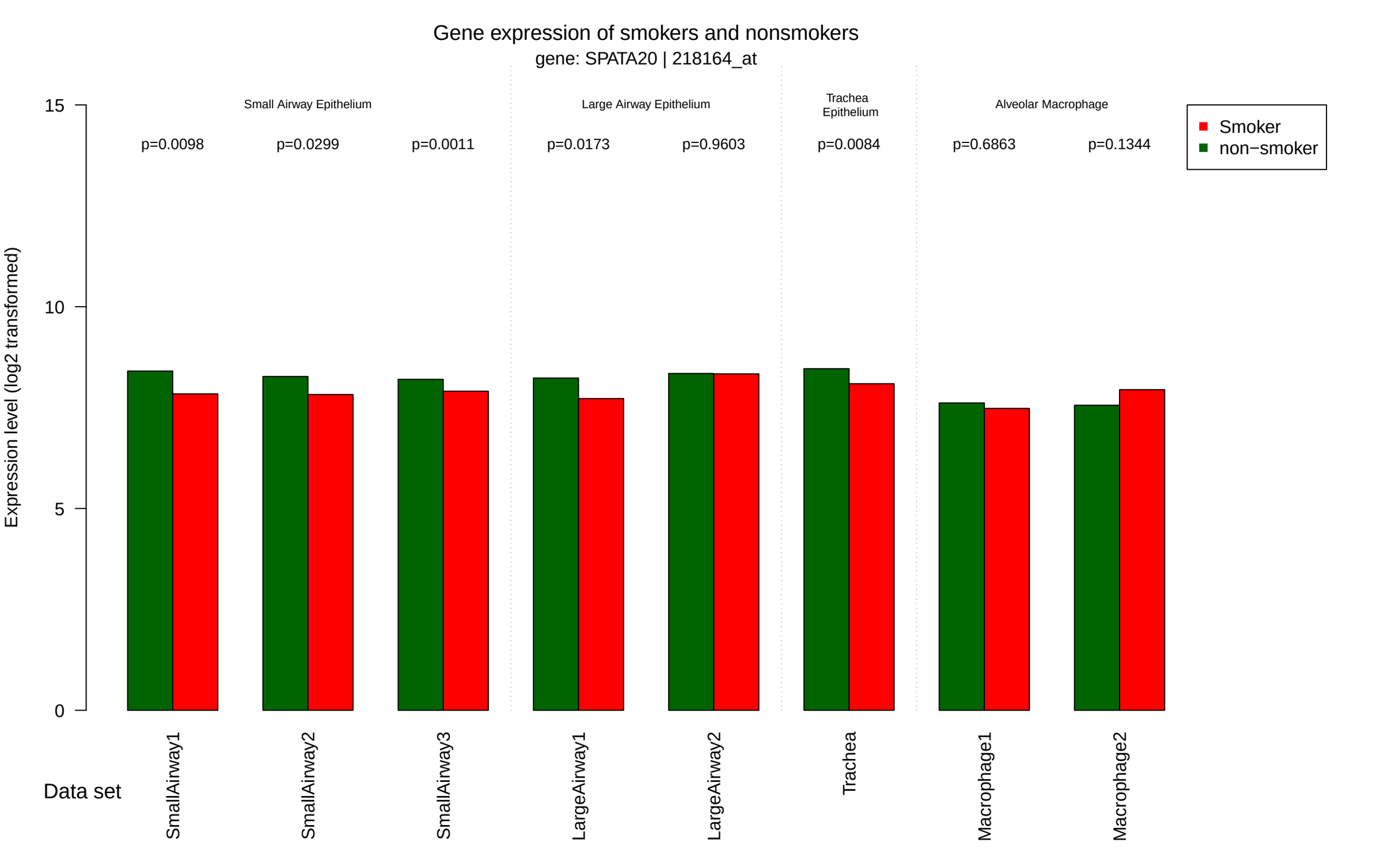
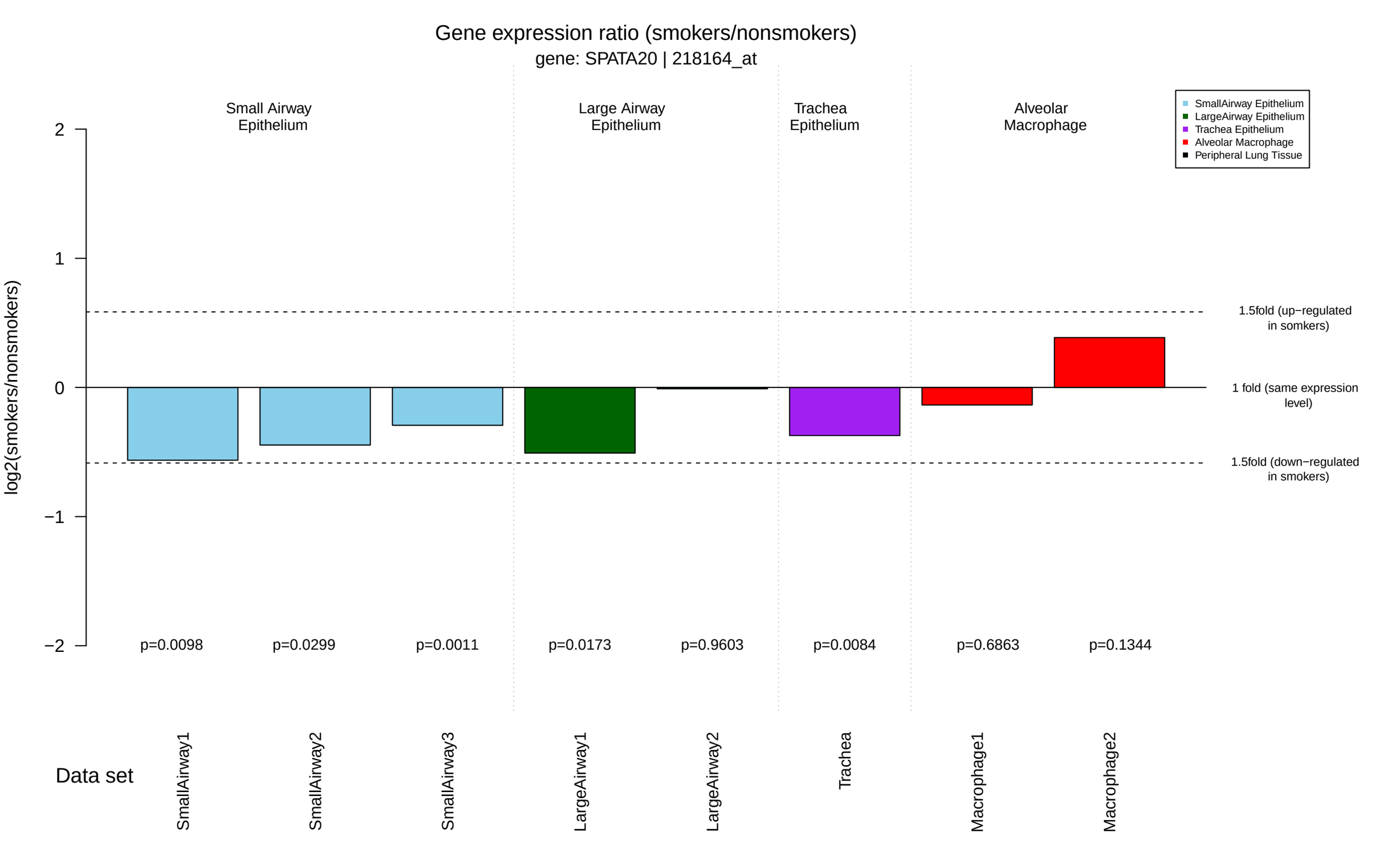
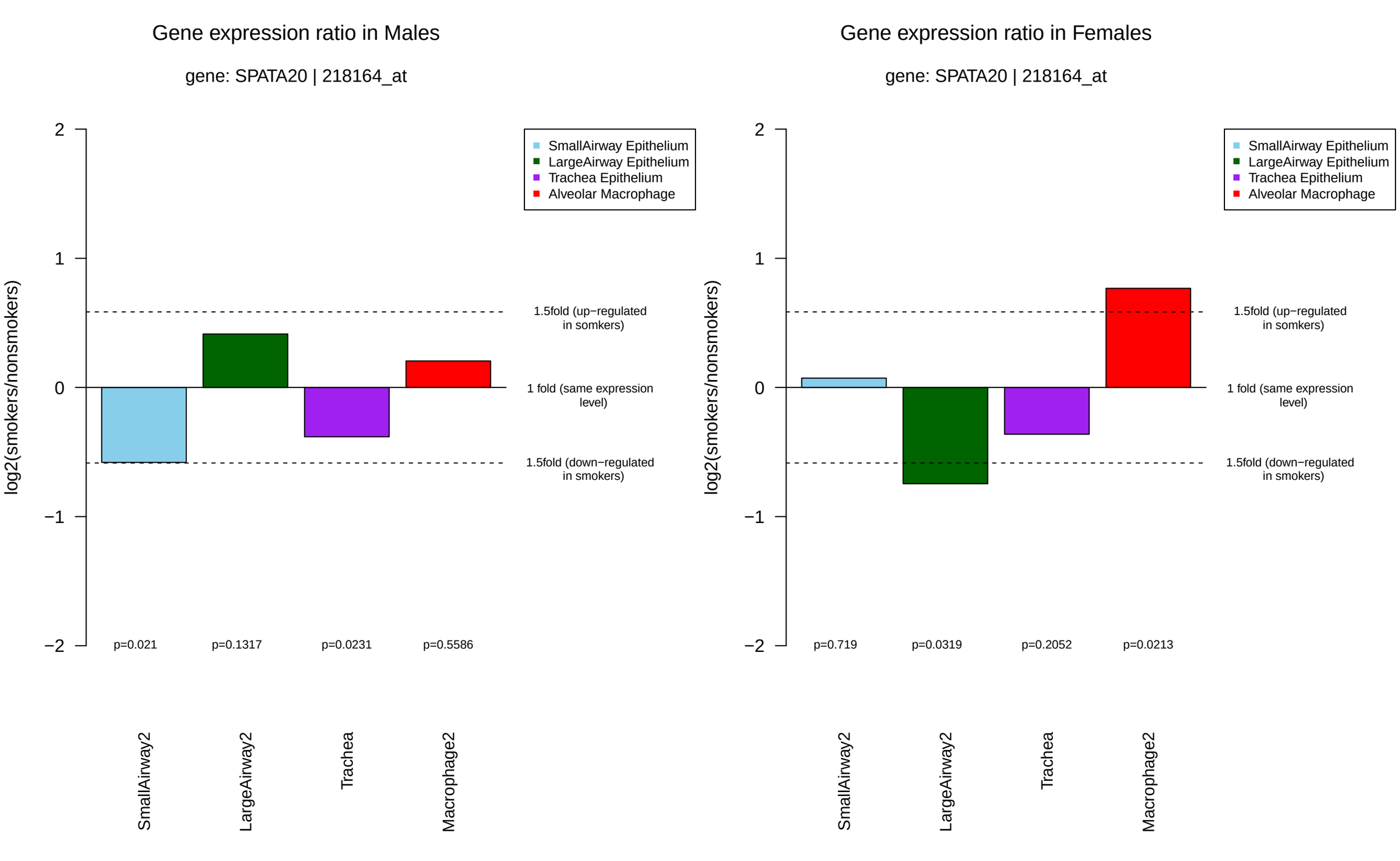
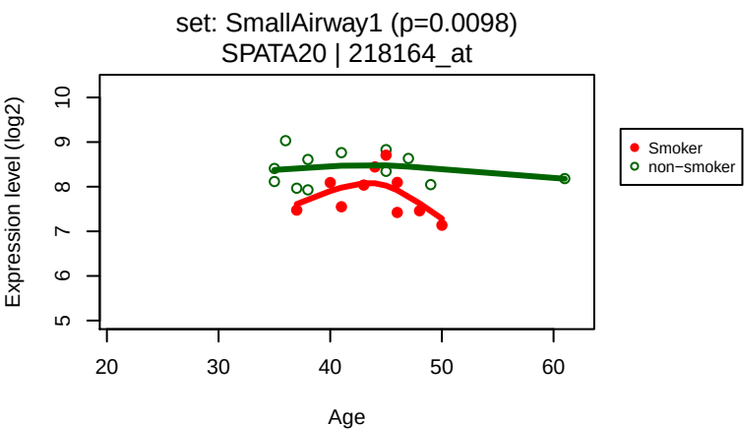
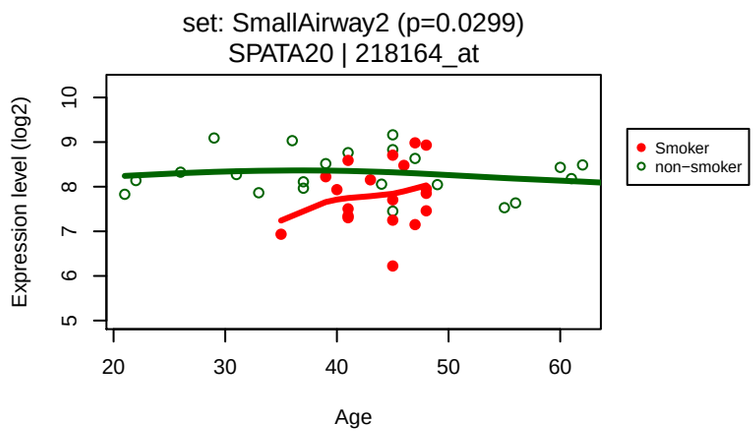
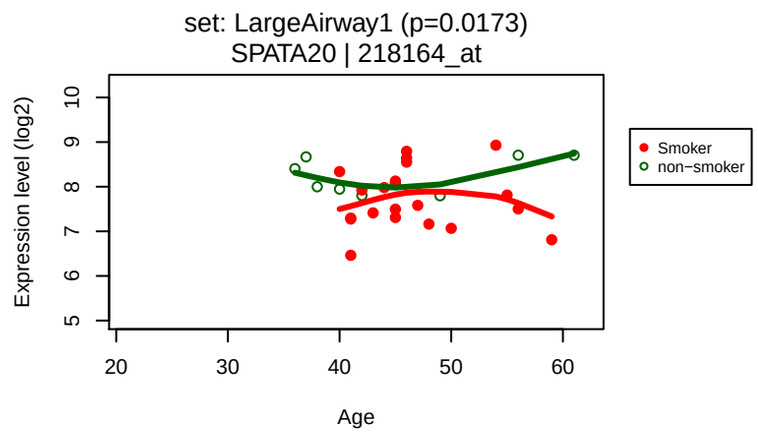

Gene expression of smokers and nonsmokers
gene: SPATA20 | 218164_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0098
p=0.0299
p=0.0011
p=0.0173
p=0.9603
p=0.0084
p=0.6863
p=0.1344
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

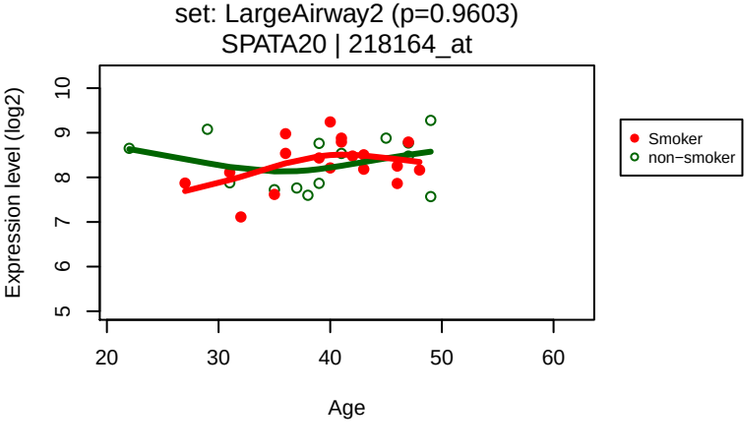

Gene expression ratio (smokers/nonsmokers)
gene: SPATA20 | 218164_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0098
p=0.0299
p=0.0011
p=0.0173
p=0.9603
p=0.0084
p=0.6863
p=0.1344
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

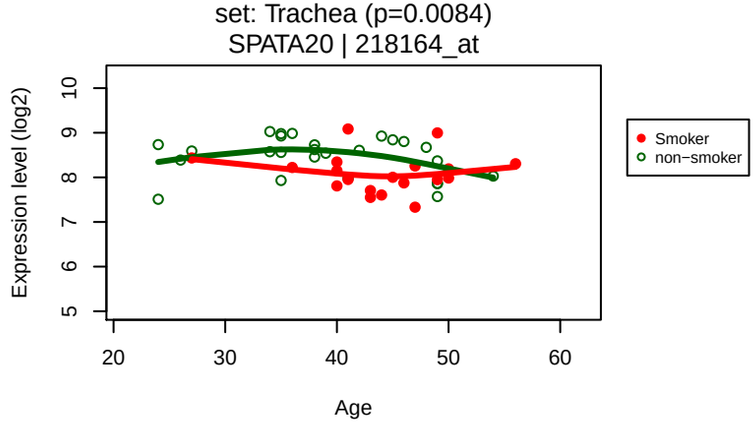

Gene expression ratio in Males
gene: SPATA20 | 218164_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.021
p=0.1317
p=0.0231
p=0.5586
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA20 | 218164_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.719
p=0.0319
p=0.2052
p=0.0213
SmallAirway2
LargeAirway2
Trachea
Macrophage2

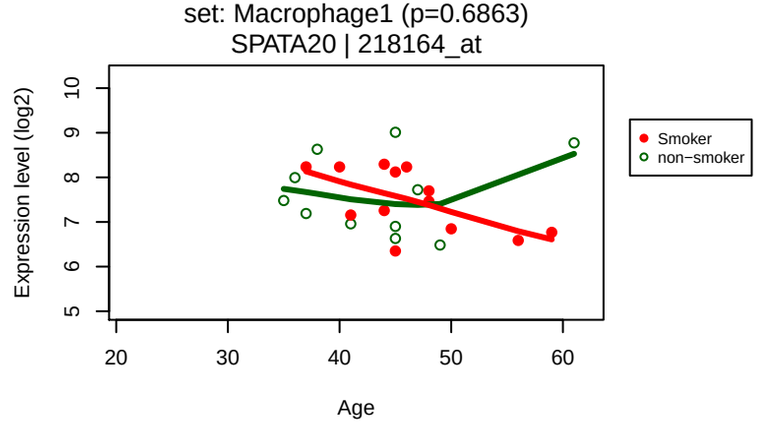

set: SmallAirway1 (p=0.0098)
SPATA20 | 218164_at
Expression level (log2)
Age
Smoker
non−smoker

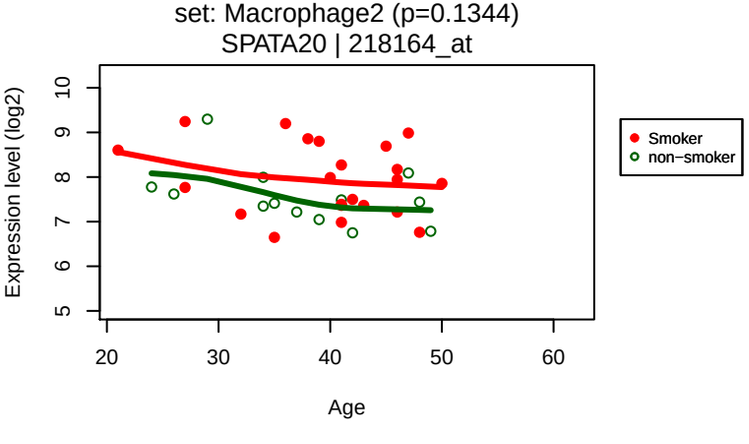

set: SmallAirway2 (p=0.0299)
SPATA20 | 218164_at
Expression level (log2)
Age
Smoker
non−smoker

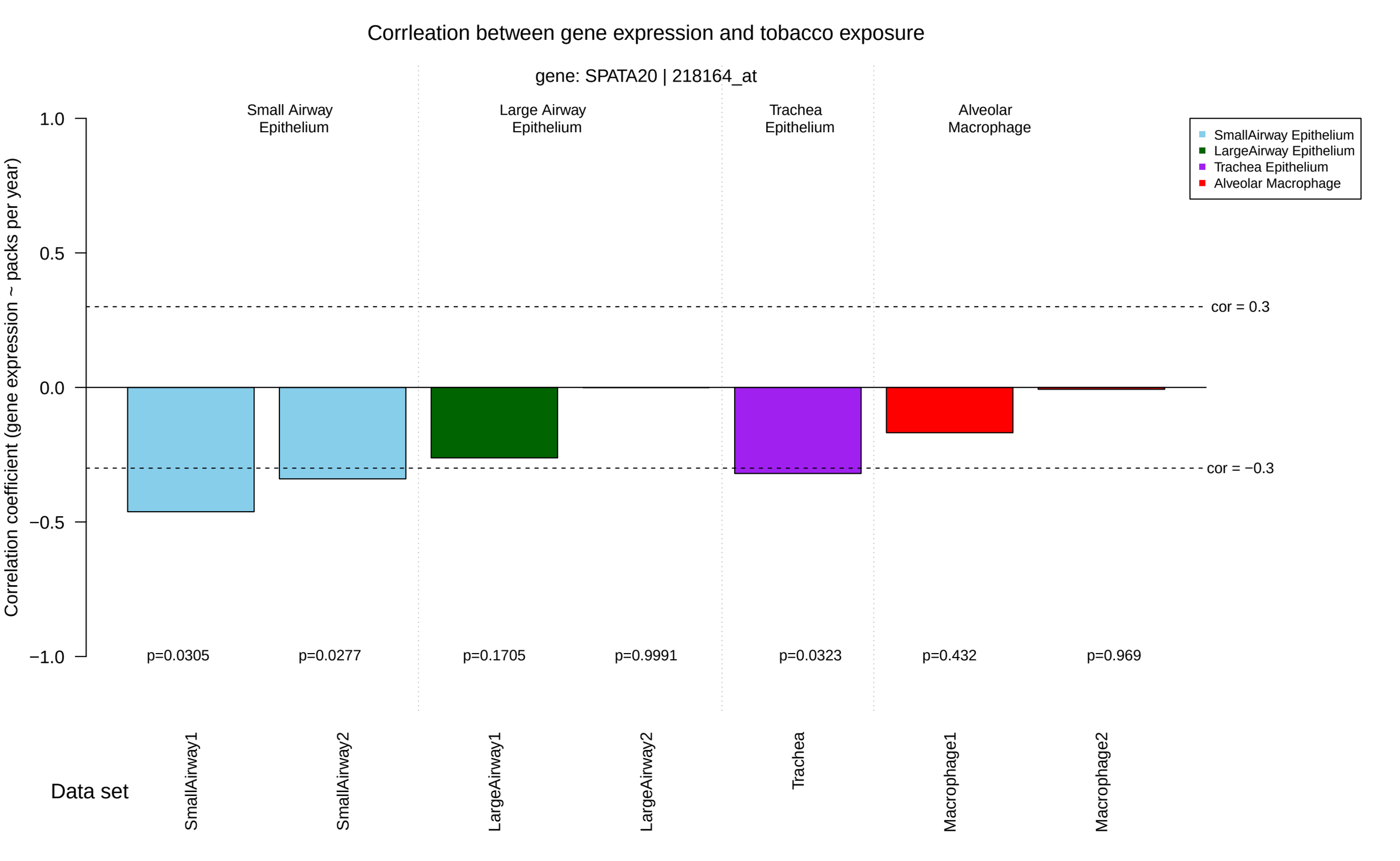

set: LargeAirway1 (p=0.0173)
SPATA20 | 218164_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9603)
SPATA20 | 218164_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0084)
SPATA20 | 218164_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6863)
SPATA20 | 218164_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1344)
SPATA20 | 218164_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA20 | 218164_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0305
p=0.0277
p=0.1705
p=0.9991
p=0.0323
p=0.432
p=0.969
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2